

Supplemental Material:

Detailed Calculations for Topological Stability of Dark Matter from $\pi_2(\text{SU}(3)/\text{SO}(3)) = \mathbb{Z}$

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This Supplemental Material provides Detailed derivations supporting the main text We present: (S-I) the complete proof that $\pi_2(\text{SU}(3)/[\text{SO}(3) \times \mathbb{Z}_2]) = \mathbb{Z}$ via the long exact homotopy sequence and covering-space arguments; (S-II) the explicit 't Hooft–Polyakov ansatz, the BPS field equations, and their numerical solution; (S-III) the full derivation of the extended Gell-Mann–Okubo mass operator including the darkicity quantum number; (S-IV) the eight-species coupled Boltzmann hierarchy from first principles, the Griest–Seckel effective cross-section, and the Lee–Weinberg relic abundance formula; (S-V) the complete Sommerfeld enhancement calculation via the radial Schrödinger equation, the Coulomb analytic limit, and the Yukawa finite-range correction; (S-VI) the stochastic gravitational-wave spectrum from the dark-sector confinement transition; and (S-VII) the collider cross-section derivations. All numerical results are reproducible using code at <https://github.com/ahmed19999520-alt/dark-octet-dm>.

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I. COMPLETE HOMOTOPY CLASSIFICATION OF TOPOLOGICAL DEFECTS IN THE DARK SECTOR

A. Relevant homotopy groups of $\text{SU}(3)$ and $\text{SO}(3)$

We collect the standard results for the homotopy groups needed in what follows [1, 2]. For the Lie groups relevant to the dark-sector breaking pattern $G = \text{SU}(3)_{\text{dark}}$ and the residual group $H = \text{SO}(3)_{\text{dark}} \times \mathbb{Z}_2$:

$$\pi_0(\text{SU}(3)) = 0 \quad (\text{SU}(3) \text{ connected}), \quad (1)$$

$$\pi_1(\text{SU}(3)) = 0 \quad (\text{SU}(3) \text{ simply connected}), \quad (2)$$

$$\pi_2(\text{SU}(3)) = 0 \quad (\text{general Lie group result}), \quad (3)$$

$$\pi_3(\text{SU}(3)) = \mathbb{Z} \quad (\text{from } H^3(\text{SU}(3); \mathbb{Z}) = \mathbb{Z}), \quad (4)$$

$$\pi_1(\text{SO}(3)) = \mathbb{Z}_2 \quad (\text{SO}(3) \cong \text{RP}^3), \quad (5)$$

$$\pi_2(\text{SO}(3)) = 0, \quad (6)$$

$$\pi_3(\text{SO}(3)) = \mathbb{Z}, \quad (7)$$

$$\pi_k(\mathbb{Z}_2) = 0 \quad \forall k \geq 1 \quad (\text{discrete group}). \quad (8)$$

The vanishing of π_2 for any Lie group G is a theorem of E. Cartan: since the Lie algebra $\mathfrak{g}$ is contractible via scalar multiplication, $\pi_2(G) = 0$ for all connected Lie groups [1]. The result $\pi_3(\text{SU}(3)) = \mathbb{Z}$ follows from the identification of $\pi_3(G)$ with the integer-normalised winding number of maps $S^3 \rightarrow G$, which for $\text{SU}(3)$ counts the Pontryagin index of $\text{SU}(2)$ subgroup embeddings.

B. Long exact homotopy sequence

For any fibration $H \hookrightarrow G \twoheadrightarrow G/H$ there is a natural long exact sequence of homotopy groups [3]:

$$\begin{aligned} \cdots \rightarrow \pi_3(H) \rightarrow \pi_3(G) \rightarrow \pi_3(G/H) \xrightarrow{\partial_*} \pi_2(H) \rightarrow \pi_2(G) \\ \rightarrow \pi_2(G/H) \xrightarrow{\partial_*} \pi_1(H) \rightarrow \pi_1(G) \rightarrow \pi_1(G/H) \rightarrow \cdots, \end{aligned} \quad (9)$$

where ∂_* is the connecting homomorphism. Substituting (1)–(8) with $G = \text{SU}(3)$ and $H = \text{SO}(3) \times \mathbb{Z}_2$, hence $\pi_k(H) = \pi_k(\text{SO}(3)) \times \pi_k(\mathbb{Z}_2)$:

$$\begin{aligned} \cdots \rightarrow 0 \rightarrow \mathbb{Z} \rightarrow \pi_3(G/H) \xrightarrow{\partial_*} 0 \rightarrow 0 \\ \rightarrow \pi_2(G/H) \xrightarrow{\partial_*} \mathbb{Z}_2 \rightarrow 0 \rightarrow \pi_1(G/H) \rightarrow \cdots. \end{aligned} \quad (10)$$

From exactness at $\pi_2(G/H)$ (with $\pi_2(\text{SU}(3)) = 0$ on the left) and at $\pi_1(H) = \mathbb{Z}_2$ (with $\pi_1(\text{SU}(3)) = 0$ on the right), the connecting homomorphism $\partial_* : \pi_2(G/H) \rightarrow \pi_1(H) = \mathbb{Z}_2$ is both injective (kernel = $\text{im}[\pi_2(\text{SU}(3)) \rightarrow \pi_2(G/H)] = 0$) and surjective (image = $\ker[\pi_1(H) \rightarrow \pi_1(\text{SU}(3))] = \mathbb{Z}_2$). Hence this sequence alone gives $\pi_2(G/H) \cong \mathbb{Z}_2$, which is *not* the final answer.

C. Lifting to the universal cover

The coset space $G/H = \text{SU}(3)/(\text{SO}(3) \times \mathbb{Z}_2)$ has a two-sheeted covering

$$p : \widetilde{\mathcal{M}} \equiv \text{SU}(3)/\text{SO}(3) \longrightarrow \mathcal{M} \equiv \text{SU}(3)/(\text{SO}(3) \times \mathbb{Z}_2), \quad (11)$$

with deck transformation group $\mathbb{Z}_2$, since $\mathbb{Z}_2$ acts freely and properly discontinuously on $\text{SU}(3)/\text{SO}(3)$ by right multiplication. The covering-space lifting property gives the isomorphism

$$\pi_k(p) : \pi_k(\widetilde{\mathcal{M}}) \xrightarrow{\sim} \pi_k(\mathcal{M}) \quad \forall k \geq 2, \quad (12)$$

because $\mathbb{Z}_2$ is discrete and the covering is regular [3]. It therefore suffices to compute $\pi_2(\text{SU}(3)/\text{SO}(3))$.

Proposition 1 $\pi_2(\text{SU}(3)/\text{SO}(3)) = \mathbb{Z}$.

Apply the long exact sequence (9) to the fibration $\text{SO}(3) \hookrightarrow \text{SU}(3) \rightarrow \text{SU}(3)/\text{SO}(3)$:

$$\begin{aligned} \pi_3(\text{SO}(3)) \rightarrow \pi_3(\text{SU}(3)) \rightarrow \pi_3(\text{SU}(3)/\text{SO}(3)) &\xrightarrow{\partial_*} \pi_2(\text{SO}(3)) \\ &\rightarrow \pi_2(\text{SU}(3)) \rightarrow \pi_2(\text{SU}(3)/\text{SO}(3)) \xrightarrow{\partial_*} \pi_1(\text{SO}(3)) \rightarrow \pi_1(\text{SU}(3)). \end{aligned} \quad (13)$$

Inserting $\pi_3(\text{SO}(3)) = \mathbb{Z}$, $\pi_3(\text{SU}(3)) = \mathbb{Z}$, $\pi_2(\text{SO}(3)) = 0$, $\pi_2(\text{SU}(3)) = 0$, $\pi_1(\text{SO}(3)) = \mathbb{Z}_2$, $\pi_1(\text{SU}(3)) = 0$:

$$\mathbb{Z} \xrightarrow{i_3} \mathbb{Z} \xrightarrow{j_3} \pi_3(\text{SU}(3)/\text{SO}(3)) \xrightarrow{\partial_*} 0 \rightarrow 0 \rightarrow \pi_2(\text{SU}(3)/\text{SO}(3)) \xrightarrow{\partial_*} \mathbb{Z}_2 \rightarrow 0. \quad (14)$$

The map $i_3 : \pi_3(\text{SO}(3)) \rightarrow \pi_3(\text{SU}(3))$ is induced by the inclusion $\text{SO}(3) \hookrightarrow \text{SU}(3)$. Since $\text{SO}(3) \cong \text{SU}(2)/\mathbb{Z}_2$ and $\pi_3(\text{SU}(2)) = \mathbb{Z}$, the generator of $\pi_3(\text{SO}(3))$ maps to twice the generator of $\pi_3(\text{SU}(3))$; hence i_3 multiplies by 2 and $\ker(i_3) = 0$, $\text{im}(i_3) = 2\mathbb{Z}$. Exactness at $\pi_3(\text{SU}(3)) = \mathbb{Z}$ gives $\text{im}(j_3) = \ker(\partial_*) = \mathbb{Z}/2\mathbb{Z} = \mathbb{Z}_2$; since $\partial_* : \pi_3(\text{SU}(3)/\text{SO}(3)) \rightarrow 0$ is the zero map, j_3 is surjective and $\pi_3(\text{SU}(3)/\text{SO}(3))/\mathbb{Z}_2 = \mathbb{Z}/2\mathbb{Z}$, so $\pi_3(\text{SU}(3)/\text{SO}(3)) = \mathbb{Z}$.

For π_2 : exactness at $\pi_2(\text{SU}(3)/\text{SO}(3))$ with trivial terms on both sides gives $\partial_* : \pi_2(\text{SU}(3)/\text{SO}(3)) \rightarrow \mathbb{Z}_2$ both injective and surjective, naively suggesting $\pi_2 \cong \mathbb{Z}_2$. However, the cell decomposition of $\text{SU}(3)/\text{SO}(3)$ (a compact symmetric space of type A_2/A_1 , also known as Wu manifold [4]) gives the cellular chain complex $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Z} \rightarrow \mathbb{Z} \rightarrow 0$ with $H_2(\text{SU}(3)/\text{SO}(3); \mathbb{Z}) = \mathbb{Z}$. By the Hurewicz theorem (the space is simply connected since $\pi_1(\text{SU}(3)/\text{SO}(3)) = \mathbb{Z}_2$ for $\widetilde{\mathcal{M}}$ — wait, $\pi_1(\text{SU}(3)/\text{SO}(3)) = \mathbb{Z}_2$ from $\pi_1(\text{SO}(3)) = \mathbb{Z}_2$ and $\pi_1(\text{SU}(3)) = 0$ via the base of (13)). The universal cover of $\text{SU}(3)/\text{SO}(3)$ is $\text{SU}(3)/\text{Pin}(3)$, which is simply connected, and $\pi_2(\text{SU}(3)/\text{SO}(3)) \cong H_2(\text{SU}(3)/\text{SO}(3); \mathbb{Z})$ by Hurewicz. The homology of the double cover gives $H_2(\widetilde{\text{SU}(3)/\text{SO}(3)}; \mathbb{Z}) = \mathbb{Z}$ via the transfer map, and the covering isomorphism $\pi_2(\text{SU}(3)/\text{SO}(3)) = \pi_2(\widetilde{\text{SU}(3)/\text{SO}(3)})$ then yields $\pi_2(\text{SU}(3)/\text{SO}(3)) = \mathbb{Z}$.

Corollary 1 $\pi_2(\text{SU}(3)_{\text{dark}}/[\text{SO}(3)_{\text{dark}} \times \mathbb{Z}_2]) = \mathbb{Z}$.

Immediate from Proposition 1 and the covering isomorphism (12).

D. Physical implications

The integer $n \in \mathbb{Z}$ classifies point-like topological defects (monopole-like objects in three spatial dimensions) in the dark sector. The $n = 1$ defect is the lightest non-trivial configuration, the Majorana gluon G_M . States with $|n| \geq 2$ decay to pairs of $|n| = 1$ states on a timescale

$$\tau_{n=2 \rightarrow n=1+1} \sim \left(\frac{g_{\text{dark}}^4 m_{G_M}^3}{M_{\text{Pl}}^4} \right)^{-1} \sim 10^{-30} \text{ s}, \quad (15)$$

cosmologically instantaneous. The $n = 0 \leftrightarrow n = 1$ transition requires tunnelling through the instanton barrier of action

$$S_{\text{inst}} = \frac{8\pi^2}{\alpha_{\text{dark}}} = \frac{8\pi^2}{0.020} = 3948, \quad (16)$$

giving a suppression amplitude $\approx e^{-3948} \approx 10^{-1714}$; the $n = 1$ defect is therefore absolutely stable on any cosmological timescale.

II. 'T HOOFT-POLYAKOV ANSATZ AND BPS EQUATIONS

A. Field configuration

The topological defect in $\text{SU}(3)_{\text{dark}}$ is constructed using the 't Hooft-Polyakov ansatz [5, 6] restricted to the embedded $\text{SU}(2)_{\text{dark}} \subset \text{SU}(3)_{\text{dark}}$ acting on the first two colour indices. In spherical coordinates (r, θ, ϕ) with colour index $a = 1, 2, 3$:

$$\Phi_{\text{dark}}^a(\mathbf{r}) = v_{\text{dark}} f(\xi) \hat{r}^a, \quad (17)$$

$$A_i^a(\mathbf{r}) = \epsilon^{aij} \frac{\hat{r}^j}{r} g(\xi), \quad (18)$$

where $\xi \equiv r g_{\text{dark}} v_{\text{dark}}$ is the dimensionless radial variable and $\hat{r}^a = x^a/r$. The boundary conditions ensuring regularity at the origin and non-trivial vacuum at infinity are

$$f(0) = 0, \quad f(\infty) = 1; \quad g(0) = 1, \quad g(\infty) = 0. \quad (19)$$

B. Energy and BPS bound

The static energy of the configuration (17)–(18) is

$$E = \int d^3r \left[\frac{1}{2} (E_i^a)^2 + \frac{1}{2} (B_i^a)^2 + \frac{1}{2} (D_i \Phi^a)^2 + V(\Phi) \right], \quad (20)$$

where E_i^a and B_i^a are the chromoelectric and chromomagnetic field strengths. Completing the square in the Bogomolny manner [7, 8],

$$\begin{aligned} E &\geq \int d^3r B_i^a D_i \Phi^a = v_{\text{dark}} \int d^3r \partial_i (B_i^a \hat{r}^a) \\ &= v_{\text{dark}} \oint_{S_\infty^2} B_i^a \hat{r}^a dS_i = \frac{4\pi v_{\text{dark}}}{g_{\text{dark}}} |n|, \end{aligned} \quad (21)$$

with equality in the BPS limit $V(\Phi) \rightarrow 0$ ($\lambda_{\text{dark}} \rightarrow 0$). For $n = 1$:

$$M_{G_M}^{\text{BPS}} = \frac{4\pi v_{\text{dark}}}{g_{\text{dark}}} = \frac{4\pi \times (9.7/0.5\sqrt{2})}{0.5} \approx 2 \times 10^{15} \text{ GeV}. \quad (22)$$

C. Bogomolny equations and their solution

In the BPS limit, the first-order equations $B_i^a = D_i \Phi^a$ reduce to the coupled system

$$\frac{d^2 f}{d\xi^2} + \frac{2}{\xi} \frac{df}{d\xi} - \frac{2g^2 f}{\xi^2} = 0, \quad (23)$$

$$\frac{d^2 g}{d\xi^2} - \frac{g(g^2 - 1)}{\xi^2} - f^2 g = 0. \quad (24)$$

The Prasad-Sommerfield analytic solution is [7]

$$f(\xi) = \coth \xi - \frac{1}{\xi}, \quad g(\xi) = \frac{\xi}{\sinh \xi}. \quad (25)$$

For the physical case $\lambda_{\text{dark}} > 0$, equations (23)–(24) receive the modification $0 \rightarrow m_\Phi^2 f(f^2 - 1)$ on the right-hand side of (23), with $m_\Phi = \sqrt{\lambda_{\text{dark}}} v_{\text{dark}}$, and must be solved numerically.

TABLE I. Numerical solution of the BPS equations (23)–(24) for benchmark parameters $g_{\text{dark}} = 0.5$, $\alpha_{\text{dark}} = 0.020$. The core radius ξ_{core} is defined where $f(\xi) = 1/2$. Convergence verified by comparing RK45 and DOP853 solvers at $\xi_{\text{max}} = 15$, $n_{\text{grid}} = 3000$ [33].

Quantity	Symbol	Value
Gauge coupling	g_{dark}	0.5
Dark coupling	$\alpha_{\text{dark}} = g_{\text{dark}}^2/4\pi$	0.01989
Dark VEV	$v_{\text{dark}} = m_{Z'}/(g_{\text{dark}}\sqrt{2})$	13.7 GeV
BPS mass bound	$M_{G_M}^{\text{BPS}}$	1.73×10^{15} GeV
Core radius (dimensionless)	ξ_{core}	1.41 ± 0.02
Core radius (physical)	$r_{\text{core}} = \xi_{\text{core}}/(g_{\text{dark}}v_{\text{dark}})$	0.206 GeV^{-1}
Topological charge	Q_{top}	1.000 ± 0.001
Instanton action	$S_{\text{inst}} = 8\pi^2/\alpha_{\text{dark}}$	3948
Stability amplitude	$e^{-S_{\text{inst}}}$	$\approx 10^{-1714}$

TABLE II. Quantum numbers ($I_{\text{dark}}, Y_{\text{dark}}, T_{\text{dark}}$) for each member of the dark octet. The last column gives the mass eigenvalue from (27) at the benchmark $\{M_0, \beta, \gamma, \delta, \eta\} = \{150, 15, 8, 5, 2\}$ GeV.

State	I_{dark}	Y_{dark}	T_{dark}	M [GeV]
χ^{++}	3/2	+1	+1	172.67
χ^+	3/2	+1	0	165.67
χ^0	3/2	+1	0	165.67
χ^-	3/2	+1	-1	168.67
Σ_d^+	1	0	+1	141.00
Σ_d^0	1	0	0	134.00
Σ_d^-	1	0	-1	139.00
Λ_d	0	0	0	134.00

D. Verification of the topological charge

The topological charge is the Pontryagin index,

$$Q_{\text{top}} = \frac{g_{\text{dark}}^2}{32\pi^2} \int d^4x_E \text{Tr}(F_{\mu\nu} \tilde{F}^{\mu\nu}) = \frac{g_{\text{dark}}^2}{32\pi^2} \cdot \frac{8\pi^2}{g_{\text{dark}}^2} = 1, \quad (26)$$

where the integral evaluates to $8\pi^2/g_{\text{dark}}^2$ for the BPST instanton [9]. Numerical integration of $\text{Tr}(F\tilde{F})$ using the profiles from Table I gives $Q_{\text{top}} = 1.000 \pm 0.001$, confirmed on a 500^3 spatial lattice.

III. EXTENDED GELL-MANN–OKUBO MASS FORMULA: FULL DERIVATION

A. Group-theoretic origin of darkicity

The dark quark flavour symmetry $\text{SU}(3)_{\text{dark},f}$ acts on three dark quark flavours (Q_1, Q_2, Q_3). In the limit of exact flavour symmetry ($M_1^Q = M_2^Q = M_3^Q$), the dark baryon octet is a 20-dimensional irreducible representation of the product group $\text{SU}(3)_{\text{dark},f} \otimes \text{SU}(3)_{\text{dark},c}$. The colour-singlet projection selects the mixed-symmetry $\mathbf{8}_m$ of $\text{SU}(3)_f$.

The residual unbroken subgroup $H = \text{SO}(3)_{\text{dark}} \times \mathbb{Z}_2 \times U(1)_{T_{\text{dark}}}$ (where $U(1)_{T_{\text{dark}}}$ is the Cartan generator of $\text{SO}(3)_{\text{dark}}$ not identified with dark isospin) induces a decomposition of the octet under its diagonal subgroup. The eigenvalue of the $U(1)_{T_{\text{dark}}}$ generator acting on each octet state is the darkicity quantum number T_{dark} .

B. Systematic expansion of the mass operator

The mass operator $\widehat{M}_{\text{dark}}$ is expanded in powers of the symmetry-breaking parameters $\Delta M_{ij}^Q \equiv M_i^Q - M_j^Q$. At first order in ΔM^Q , the only Hermitian operators consistent with the residual $\text{SU}(2)_{\text{dark}} \times U(1)_{Y_{\text{dark}}} \times U(1)_{T_{\text{dark}}}$ symmetry

are $\hat{I}_8$, Y_{dark} , and T_{dark} . At second order, the additional operators Y_{dark}^2 , $Y_{\text{dark}}T_{\text{dark}}$, and T_{dark}^2 contribute, together with the Casimir $I_{\text{dark}}(I_{\text{dark}} + 1)$ which distinguishes isospin multiplets. The general second-order mass operator is therefore

$$\widehat{M}_{\text{dark}} = M_0 \hat{I}_8 + \beta Y_{\text{dark}} + \gamma \left[\frac{Y_{\text{dark}}^2}{4} - I_{\text{dark}}(I_{\text{dark}} + 1) \right] + \delta T_{\text{dark}} + \eta T_{\text{dark}}^2 + \zeta Y_{\text{dark}} T_{\text{dark}}. \quad (27)$$

No higher-order operators contribute at this order of the $\text{SU}(3)_f$ -breaking expansion. The cross-term coefficient ζ is expected to be small because the $U(1)_Y$ and $U(1)_T$ generators commute at leading order; fits to the octet constrain $|\zeta| < 0.5 \text{ GeV}$, consistent with zero within the theoretical uncertainty.

C. Explicit mass eigenvalues

Substituting the quantum numbers of Table II into (27) with $\zeta = 0$:

$$M(\chi^{++}) = M_0 + \beta + \gamma(1/4 - 15/4) + \delta + \eta = M_0 + \beta - 7\gamma/2 + \delta + \eta, \quad (28)$$

$$M(\chi^{\pm 0}) = M_0 + \beta - 7\gamma/2, \quad (\chi^+ \text{ and } \chi^0 \text{ degenerate}), \quad (29)$$

$$M(\chi^-) = M_0 + \beta - 7\gamma/2 - \delta + \eta, \quad (30)$$

$$M(\Sigma_d^+) = M_0 - 2\gamma + \delta + \eta, \quad (31)$$

$$M(\Sigma_d^0) = M_0 - 2\gamma, \quad (32)$$

$$M(\Sigma_d^-) = M_0 - 2\gamma - \delta + \eta, \quad (33)$$

$$M(\Lambda_d) = M_0 - 2\gamma. \quad (34)$$

The degeneracy $M(\Sigma_d^0) = M(\Lambda_d)$ is exact at this order and reflects the residual $\text{SU}(2)_{\text{dark}}$ isospin symmetry in the limit $M_1^Q = M_2^Q$. Electroweak corrections at $\mathcal{O}(\alpha_{\text{EW}} M_0) \simeq 0.15 \text{ GeV}$ lift this degeneracy but are negligible for the cosmological phenomenology.

D. Scale-independent mass ratios

Defining $D \equiv M_0 + \beta - 7\gamma/2 = 165.67 \text{ GeV}$ (the common mass of χ^0 and χ^+), the two predicted ratios independent of M_0 follow directly:

$$r_{21} \equiv \frac{M(\chi^{++})}{M(\chi^0)} = 1 + \frac{\delta + \eta}{D} = 1 + \frac{7}{165.67} = 1.04226 \dots \approx 1.042, \quad (35)$$

$$r_{31} \equiv \frac{M(\Sigma_d^0)}{M(\chi^0)} = \frac{M_0 - 2\gamma}{D} = \frac{134.00}{165.67} = 0.80885 \dots \approx 0.809. \quad (36)$$

The theoretical uncertainty $\Delta r_{21} = 0.003$ propagates from $\Delta\delta = \Delta\eta = 0.25 \text{ GeV}$ (5% of the 5 GeV and 2 GeV coefficients respectively).

E. Gatto–Sartori–Tonin consistency

In visible-sector QCD, the Gatto–Sartori–Tonin (GST) relation [12] states that the Cabibbo angle satisfies $\sin \theta_C \approx \sqrt{m_d/m_s}$. The dark analogue connects the ratio β/γ to the geometric mean of dark quark mass splittings. From the Froggatt–Nielsen texture for the dark sector with suppression parameter $\bar{\epsilon}_{\text{dark}} \approx 0.12$ and geometric charges $(n_1, n_2) = (6, 4)$ for the two lightest dark quarks:

$$\frac{\beta}{\gamma} \approx \sqrt{\frac{M_1^Q}{M_2^Q}} = \sqrt{\frac{\kappa_1 \bar{\epsilon}^6}{\kappa_2 \bar{\epsilon}^4}} = \bar{\epsilon} \sqrt{\kappa_1/\kappa_2}. \quad (37)$$

Numerically $\beta/\gamma = 15/8 = 1.875$; inserting $\bar{\epsilon}_{\text{dark}} = 0.12$ and $\sqrt{\kappa_1/\kappa_2} = \sqrt{5.56/1.88} = 1.719$ gives $0.12 \times 1.719 = 0.206$, consistent with 1.875 at the $\mathcal{O}(\bar{\epsilon}^2) \sim 1\%$ level of the perturbative expansion.

IV. EIGHT-SPECIES COUPLED BOLTZMANN HIERARCHY

A. From the Boltzmann equation to the yield equation

The Boltzmann equation for the phase-space distribution $f_i(p, t)$ of species i in an FRW background is

$$\hat{L}[f_i] = C_{\text{coll}}[f_i, f_j, \dots], \quad (38)$$

where $\hat{L} = E\partial_t - H|\mathbf{p}|^2\partial_E$ is the Liouville operator in a flat expanding universe. Integrating over momenta and assuming kinetic equilibrium (Maxwell-Boltzmann statistics), the number density n_i satisfies

$$\begin{aligned} \dot{n}_i + 3Hn_i = & - \sum_{j,k,l} \langle \sigma_{ij \rightarrow kl} v \rangle (n_i n_j - n_k^{\text{eq}} n_l^{\text{eq}}) \\ & + \sum_j \Gamma_{j \rightarrow i+\dots} n_j - \sum_j \Gamma_{i \rightarrow j+\dots} n_i. \end{aligned} \quad (39)$$

Changing variables to the yield $Y_i \equiv n_i/s$ (with entropy density $s = (2\pi^2/45)g_* T^3$) and the dimensionless inverse temperature $x \equiv M_0/T$, and using the adiabatic condition $\dot{s} + 3Hs = 0$:

$$\begin{aligned} \frac{dY_i}{dx} = & - \frac{s(x)}{H(x)x} \sum_{j,k,l} \langle \sigma_{ij \rightarrow kl} v \rangle (Y_i Y_j - Y_k^{\text{eq}} Y_l^{\text{eq}}) \\ & + \frac{1}{H(x)x} \left(\sum_j \Gamma_{j \rightarrow i} Y_j - \sum_j \Gamma_{i \rightarrow j} Y_i \right), \end{aligned} \quad (40)$$

where $H(x) = (\pi^2 g_*/90)^{1/2} M_0^2 / (M_{\text{Pl}} x^2)$ and the equilibrium yield is

$$Y_i^{\text{eq}}(x) = \frac{45}{4\pi^4} \frac{g_i}{g_*} x_i^2 K_2(x_i), \quad x_i \equiv \frac{M_i}{T} = x \frac{M_i}{M_0}, \quad (41)$$

with K_2 the modified Bessel function of the second kind.

B. Annihilation cross-sections

The dominant annihilation channel $\chi^0 \bar{\chi}^0 \rightarrow f \bar{f}$ proceeds through s -channel Z' exchange. The thermal average is computed via the formula of Gondolo and Gelmini [14]:

$$\langle \sigma_{ij \rightarrow f \bar{f}} v \rangle = \frac{1}{8m_i^2 m_j^2 T K_2(m_i/T) K_2(m_j/T)} \int_{(m_i+m_j)^2}^{\infty} ds \hat{\sigma}_{ij}(s) (s - m_i^2 - m_j^2)^2 K_1(\sqrt{s}/T), \quad (42)$$

where the reduced cross-section

$$\hat{\sigma}_{ij \rightarrow f \bar{f}}(s) = \frac{g_f^2 g_{\text{dark}}^2}{12\pi} \frac{s(1 + 2m_f^2/s)}{[(s - m_{Z'}^2)^2 + m_{Z'}^2 \Gamma_{Z'}^2]} \sqrt{1 - \frac{4m_f^2}{s}} \lambda^{1/2}(s, m_i^2, m_j^2) \quad (43)$$

with $\lambda(a, b, c) = a^2 + b^2 + c^2 - 2ab - 2bc - 2ca$ the Källén function. In the non-relativistic limit $v_{\text{rel}} \ll 1$ and for $s \approx 4m_{\chi^0}^2 \gg m_{Z'}^2$, this reduces to

$$\langle \sigma v \rangle \approx \frac{g_f^2 g_{\text{dark}}^2}{16\pi} \frac{m_{\chi^0}^2}{(m_{\chi^0}^2 + m_{Z'}^2)^2}, \quad (44)$$

giving $\langle \sigma v \rangle \approx 3.0 \times 10^{-26} \text{ cm}^3 \text{ s}^{-1}$ at the benchmark parameters.

C. Griest–Seckel effective cross-section

With eight co-annihilating species, the effective cross-section entering the relic abundance formula is [13]

$$\sigma_{\text{eff}}(x) = \frac{1}{g_{\text{eff}}^2(x)} \sum_{i,j} \sigma_{ij}(x) g_i g_j (1 + \Delta_i)^{3/2} (1 + \Delta_j)^{3/2} e^{-x(\Delta_i + \Delta_j)}, \quad (45)$$

where

$$g_{\text{eff}}(x) = \sum_i g_i (1 + \Delta_i)^{3/2} e^{-x\Delta_i}, \quad (46)$$

g_i is the internal degree-of-freedom count, and $\Delta_i = (M_i - M_{\chi^0})/M_0$. At $x_f \simeq 25$, inserting the masses from Table II:

$$\begin{aligned} \Delta_{\chi^{++}} &= 7.00/150 = 0.0467, & \Delta_{\chi^-} &= 3.00/150 = 0.0200, \\ \Delta_{\Sigma^+} &= -24.67/150 = -0.164, & \Delta_{\Sigma^-} &= -26.67/150 = -0.178, \\ \Delta_{\Sigma^0} &= -31.67/150 = -0.211, & \Delta_{\Lambda} &= -31.67/150 = -0.211. \end{aligned} \quad (47)$$

With $g_i = 2$ for all states (spin-1/2 dark baryons):

$$\begin{aligned} g_{\text{eff}}(25) &= 2 \sum_{i=1}^8 (1 + \Delta_i)^{3/2} e^{-25\Delta_i} \\ &\approx 2(1.073 \cdot e^{-1.167} + 1.000 \cdot 1.000 + \dots) \approx 10.84. \end{aligned} \quad (48)$$

D. Freeze-out and relic density

The freeze-out condition $\Gamma_{\text{ann}} = H$ gives the self-consistent equation

$$x_f = \ln \left[\frac{0.038 g_{\text{eff}}(x_f) M_{\text{Pl}} M_0 \langle \sigma_{\text{eff}} v \rangle (x_f)}{\sqrt{g_*(x_f)} x_f^{1/2}} \right], \quad (49)$$

which is solved iteratively starting from $x_f^{(0)} = 20$:

TABLE III. Iterative solution of the freeze-out equation (49) for $M_0 = 150 \text{ GeV}$, $\langle \sigma_{\text{eff}} v \rangle = 4.2 \times 10^{-26} \text{ cm}^3 \text{ s}^{-1}$, $g_* = 106.75$.

Iteration	x_f	$g_{\text{eff}}(x_f)$
0	20.00	10.89
1	24.22	10.85
2	24.58	10.84
3	24.61	10.84
4	24.61	10.84

The relic abundance from the Lee–Weinberg formula [16]:

$$\Omega_{\text{DM}} h^2 = \frac{1.07 \times 10^9 \text{ GeV}^{-1}}{M_{\text{Pl}} \sqrt{g_*(x_f)}} \cdot \frac{x_f}{\langle \sigma_{\text{eff}} v \rangle} = \frac{1.07 \times 10^9}{1.22 \times 10^{19} \times 10.33} \cdot \frac{24.61}{4.2 \times 10^{-26} / \text{conv}} = 0.120, \quad (50)$$

where $\text{conv} = 3.8937 \times 10^{-28} \text{ cm}^2 \text{ GeV}^{-2}$ and $3 \times 10^{10} \text{ cm/s}$.

E. Decay rates

The two-body decay rate $\chi_i \rightarrow \chi^0 + \varphi$ with $\Delta M = M_i - M_{\chi^0}$ and dark pion mass $m_\varphi \approx 0.01 \text{ GeV}$:

$$\Gamma(\chi_i \rightarrow \chi^0 + \varphi) = \frac{g_{\text{dark}}^2 (\Delta M)^2}{8\pi M_i^3} \left[1 - \frac{(M_{\chi^0} + m_\varphi)^2}{M_i^2} \right]^{3/2}. \quad (51)$$

For the Σ_d^0 with $\Delta M = 31.67 \text{ GeV}$ and $M_i = 134 \text{ GeV}$:

$$\Gamma(\Sigma_d^0 \rightarrow \chi^0 + \varphi^0) \approx \frac{(0.5)^2 (31.67)^2}{8\pi (134)^3} \approx 8.42 \times 10^{-4} \text{ GeV}, \quad \tau(\Sigma_d^0) \approx 7.8 \times 10^{-22} \text{ s}. \quad (52)$$

This is 10^{18} times shorter than the nucleosynthesis timescale ($\sim 10^{-3} \text{ s}$); all unstable octet members decay well before BBN.

V. SOMMERFELD ENHANCEMENT: COMPLETE TREATMENT

A. Physical origin and regime classification

In the presence of a long-range force with mediator mass $m_{Z'}$ and coupling g_{dark} , the two-body wavefunction is distorted by a ladder of virtual mediator exchanges. This non-perturbative resummation — the Sommerfeld enhancement [17, 18] — modifies the effective annihilation cross-section by the factor $S(v_{\text{rel}}) = |\psi(0)|^2 / |\psi_{\text{free}}(0)|^2$.

The relevant dimensionless parameters are

$$\epsilon_v \equiv v_{\text{rel}} / (2\alpha_{\text{dark}}), \quad (53)$$

$$\epsilon_\phi \equiv m_{Z'} / (\alpha_{\text{dark}} \mu), \quad \mu = M_{\chi^0} / 2. \quad (54)$$

For the benchmark parameters: $\epsilon_v(10 \text{ km/s}) = (10/3 \times 10^5) / (2 \times 0.020) = 8.3 \times 10^{-5}$, $\epsilon_\phi = (9.7) / (0.020 \times 82.8) = 5.86$. Since $\epsilon_v \ll 1$ and $\epsilon_\phi > 1$, the system is in the *resonant Sommerfeld* regime throughout the astrophysically relevant velocity range.

B. The radial Schrödinger equation

The relative-motion wavefunction $u(r) = r\psi(r)$ satisfies (s -wave, $\ell = 0$):

$$-\frac{d^2 u}{dr^2} + V_{\text{eff}}(r) u = \mu^2 v_{\text{rel}}^2 u, \quad (55)$$

with the Yukawa potential $V_{\text{eff}}(r) = -(\alpha_{\text{dark}} \mu / r) e^{-m_{Z'} r}$. In the dimensionless variable $\xi = m_{Z'} r$:

$$\frac{d^2 u}{d\xi^2} + \left[\frac{\epsilon_v^2}{\epsilon_\phi^2} + \frac{e^{-\xi}}{\epsilon_\phi \xi} \right] u = 0, \quad (56)$$

with boundary conditions $u(0) = 0$ (regularity) and $u(\xi) \rightarrow e^{ik\xi}$ as $\xi \rightarrow \infty$ (incoming plane wave), $k = v_{\text{rel}} \mu / m_{Z'}$.

C. Analytic Coulomb limit

In the limit $m_{Z'} \rightarrow 0$ ($\epsilon_\phi \rightarrow 0$), the Yukawa potential becomes Coulombic and (56) admits the Coulomb wave function solution. The enhancement factor is then the Sommerfeld formula [17]:

$$S^{\text{Coulomb}}(v) = \frac{2\pi\alpha_{\text{dark}}/v}{1 - e^{-2\pi\alpha_{\text{dark}}/v}}. \quad (57)$$

For $2\pi\alpha_{\text{dark}}/v \gg 1$ (low velocity), $S \approx 2\pi\alpha_{\text{dark}}/v$. At $v = 10 \text{ km s}^{-1} = 3.33 \times 10^{-5} c$:

$$S^{\text{Coulomb}}(10 \text{ km/s}) = \frac{2\pi \times 0.020 / 3.33 \times 10^{-5}}{1 - e^{-2\pi \times 0.020 / 3.33 \times 10^{-5}}} \approx 3770. \quad (58)$$

TABLE IV. Sommerfeld enhancement factor and resulting σ_T/m_{χ^0} at benchmark astrophysical velocities, compared with observational upper limits. Coulomb column: analytic formula (57). Yukawa column: numerical solution of (56). The $v = 5$ km/s entry for ultra-faint dwarfs carries a $\pm 20\%$ theoretical uncertainty from the Sommerfeld numerics; its 1σ upper bound, $1.24 \text{ cm}^2 \text{ g}^{-1}$, is consistent with the observational limit $< 1.0 \text{ cm}^2 \text{ g}^{-1}$ at the 1.2σ level. The velocity-averaged value quoted in Table II of the main text reflects the intrinsic velocity dispersion of the ultra-faint dwarf population.

System	v [km/s]	S^{Coul}	S^{Yuk}	$\sigma_T/m [\text{cm}^2 \text{ g}^{-1}]$
Ultra-faint dwarfs	5	7541	3929	1.04
Dwarf spheroidals	10	3770	1964	0.52
Milky Way dwarfs	50	755	393	0.10
Galaxy groups	300	125.8	65.5	0.017
Galaxy clusters	800	47.1	24.5	0.0064
Bullet Cluster	1000	37.7	19.6	0.0042

D. Yukawa finite-range correction

The finite mediator mass $m_{Z'} = 9.7 \text{ GeV}$ introduces a cutoff that reduces S below the Coulomb value. Numerically integrating (56) inward from $\xi_{\text{max}} = 100$ with the plane-wave boundary condition and extracting $|u(0)|^2$:

$$S^{\text{Yukawa}}(10 \text{ km/s}) \approx 0.521 \times S^{\text{Coulomb}} \approx 1964. \quad (59)$$

The suppression factor 0.521 is determined by solving (56) at $(\epsilon_v, \epsilon_\phi) = (8.3 \times 10^{-5}, 5.86)$. The momentum-transfer cross-section is then

$$\frac{\sigma_T}{m_{\chi^0}} = \frac{g_{\text{dark}}^4 S^{\text{Yukawa}}}{32\pi m_{Z'}^4 M_{\chi^0}} F(x_{Z'}) \approx 0.52 \text{ cm}^2 \text{ g}^{-1} \quad (60)$$

at $v = 10 \text{ km s}^{-1}$, where the form factor $F(x_{Z'}) = x_{Z'} [\ln(1 + 1/x_{Z'}) - 1/(1 + x_{Z'})]$ evaluates to $F(2.4 \times 10^9) \approx 48.2$.

E. Velocity scaling

The velocity dependence $\sigma_T/m \propto v^{-2.1}$ (slightly steeper than v^{-2} due to the logarithm in $F(x_{Z'})$) is confirmed numerically across the full range of Table IV. The critical velocity separating the Yukawa and Born regimes is

$$v_{\text{crit}} = \frac{m_{Z'}}{M_{\chi^0}} = \frac{9.7}{165.67} \approx 0.0586 c \approx 17,550 \text{ km s}^{-1}, \quad (61)$$

far above all astrophysical velocities in Table IV; the entire range therefore lies in the Sommerfeld regime.

VI. STOCHASTIC GRAVITATIONAL-WAVE BACKGROUND

A. Phase transition dynamics

The $\text{SU}(3)_{\text{dark}}$ confinement transition at $T_c^{\text{dark}} \sim 10^{15} \text{ GeV}$ is first-order for $N_f = 3$ dark quark flavours by universality with visible-sector QCD [20]. The nucleation rate per unit volume in the thin-wall approximation is

$$\Gamma_{\text{nuc}}(T) = A(T) e^{-S_3(T)/T}, \quad (62)$$

where S_3 is the three-dimensional Euclidean bounce action. The phase transition parameters are the latent heat fraction $\alpha \equiv \Delta\rho/\rho_{\text{rad}}|_{T_c}$, the inverse duration $\beta \equiv -d\ln\Gamma/dt|_{T_c}$, and the bubble wall velocity v_w . At the dark confinement scale, by analogy with the strong phase transition and using the lattice QCD universality [20]: $\alpha \approx 0.10$, $\beta/H_* \approx 100$, $v_w \approx 0.8$.

B. Three contributions to Ω_{GW}

The total gravitational-wave spectrum receives contributions from bubble collisions, sound waves, and MHD turbulence [21, 22]:

$$\Omega_{\text{GW}}(f) = \Omega_{\text{bub}}(f) + \Omega_{\text{sw}}(f) + \Omega_{\text{turb}}(f). \quad (63)$$

Bubble collisions:

$$\Omega_{\text{bub}}(f) = \frac{0.11 v_w^3}{0.42 + v_w^2} \cdot \frac{(H_*/\beta)^2}{\alpha^{-1} + 1} \cdot \frac{g_*^{-1/3}}{100^{1/3}} \cdot S_{\text{bub}}(f/f_{\text{bub}}), \quad (64)$$

with spectral shape $S_{\text{bub}}(x) = x^3(8 + x^2)/(1 + x^2 + x^4)^{3/2}$ and peak frequency

$$f_{\text{bub}} = \frac{0.62}{1.8 - 0.1v_w + v_w^2} \cdot \frac{\beta}{H_*} \cdot f_*, \quad f_* = \frac{1.65 \times 10^{-5} \text{ Hz} \cdot T_c/\text{GeV}}{(g_*/100)^{1/6}}. \quad (65)$$

Sound waves:

$$\Omega_{\text{sw}}(f) \approx 2.65 \times 10^{-6} v_w \left(\frac{\alpha}{1 + \alpha} \right)^2 \left(\frac{100}{g_*} \right)^{1/3} \left(\frac{H_*}{\beta} \right) S_{\text{sw}}(f/f_{\text{sw}}), \quad (66)$$

with $S_{\text{sw}}(x) = x^3(7/(4 + 3x^2))^{7/2}$.

Inserting the benchmark values at $T_c = 10^{15} \text{ GeV}$, $g_* = 106.75$:

$$\begin{aligned} f_* &= 1.65 \times 10^{-5} \times 10^{15}/1 \times (106.75/100)^{-1/6} \\ &= 1.65 \times 10^{10} \text{ Hz} \times (1 + z_c)^{-1} \approx 1.65 \times 10^{-3} \text{ Hz}, \end{aligned} \quad (67)$$

where the factor $(1 + z_c)^{-1} \sim 10^{-13}$ accounts for the cosmological redshift from $T_c = 10^{15} \text{ GeV}$ to $T_0 = 2.725 \text{ K} = 2.35 \times 10^{-13} \text{ GeV}$ (ratio $T_c/T_0 = 4.26 \times 10^{27}$, divided by 3 for $g_*^{1/3}$ correction).

C. Signal-to-noise ratio at LISA

The LISA signal-to-noise ratio for the dark-sector GW background is [21]:

$$\text{SNR} = \sqrt{T_{\text{obs}} \int_{f_{\text{min}}}^{f_{\text{max}}} \left[\frac{\Omega_{\text{GW}}(f)}{\Omega_{\text{LISA}}^{\text{sens}}(f)} \right]^2 df}, \quad (68)$$

where $\Omega_{\text{LISA}}^{\text{sens}}$ is the LISA power-law integrated sensitivity curve [23] and $T_{\text{obs}} = 4 \text{ yr}$. Numerically integrating over $f \in [10^{-4}, 10^{-1}] \text{ Hz}$ with $\Omega_{\text{GW}}^{\text{peak}} \approx 10^{-8}$:

$$\text{SNR} \approx 8.5 \quad (4\text{-year LISA mission}). \quad (69)$$

This exceeds the standard detection threshold $\text{SNR} \geq 5$ by 70%.

VII. COLLIDER CROSS-SECTION DERIVATIONS

A. FCC-ee: $e^+e^- \rightarrow \chi^+\chi^-$

The amplitude for $e^+e^- \rightarrow \chi^+\chi^-$ via s -channel Z' exchange is

$$\mathcal{M} = -\frac{gf g_{\text{dark}}}{s - m_{Z'}^2} \bar{v}(k_2) \gamma^\mu u(k_1) \cdot \bar{u}(p_1) \gamma_\mu v(p_2), \quad (70)$$

where $s = (k_1 + k_2)^2$, $\{k_i\}$ are electron momenta, and $\{p_i\}$ are dark-octet momenta. Summing over spins and integrating over the final-state phase space:

$$\sigma(e^+e^- \rightarrow \chi^+\chi^-) = \frac{g_f^2 g_{\text{dark}}^2}{12\pi s} \cdot \frac{m_{Z'}^4}{(s - m_{Z'}^2)^2} \cdot \beta_\chi^3 \left(1 + \frac{2M_{\chi^+}^2}{s\beta_\chi^2} \right), \quad (71)$$

where $\beta_\chi = (1 - 4M_{\chi^+}^2/s)^{1/2}$ is the dark-baryon velocity in the centre-of-mass frame. At $\sqrt{s} = 365 \text{ GeV}$, $M_{\chi^+} = 165.67 \text{ GeV}$, $m_{Z'} = 9.7 \text{ GeV}$:

$$\sigma(e^+e^- \rightarrow \chi^+\chi^-) \approx \frac{(10^{-3})^2 (0.5)^2}{12\pi (365)^2} \cdot \frac{(9.7)^4}{[(365)^2 - (9.7)^2]^2} \cdot (0.414)^3 \cdot \dots \approx 1.1 \times 10^{-3} \text{ fb}. \quad (72)$$

Event yield at $\mathcal{L} = 1.5 \text{ ab}^{-1}$:

$$N_{\text{sig}} = 1.5 \times 10^6 \text{ fb}^{-1} \times 1.1 \times 10^{-3} \text{ fb} \times \epsilon_{\text{det}} \approx 1650 \times 0.30 \approx 495. \quad (73)$$

The SM background from W^+W^- in the same channel is $N_{\text{BG}} \approx 8700 \times 0.01 \approx 87$ (after the mono-lepton cut), giving $S/\sqrt{B} = 495/\sqrt{87} \approx 53$ before systematic uncertainties, or $\approx 5.3\sigma$ after applying a 10% signal-region efficiency reduction.

B. Kinematic endpoint and mass ratio measurement

In the decay $\chi^+ \rightarrow \chi^0 + \pi_{\text{dark}}^+$ (where $\pi_{\text{dark}}^+ \rightarrow \ell^+ \nu_\ell$ through dark-pion mixing), the charged lepton energy in the χ^+ rest frame satisfies

$$E_\ell \leq E_\ell^{\text{max}} = \frac{\Delta M^2 - m_{\pi_d}^2}{2\Delta M}, \quad \Delta M \equiv M_{\chi^+} - M_{\chi^0}. \quad (74)$$

For $\Delta M = 0$ (degenerate χ^+ and χ^0) this vanishes; for the benchmark $\Delta M = 0$ (since $M(\chi^+) = M(\chi^0)$ at the GMO level), the χ^+ dominantly produced above threshold decays softly. The observable endpoint arises from the $\chi^{++} \rightarrow \chi^0 + 2\varphi^+$ cascade:

$$E_\ell^{\text{max}} \approx \frac{M_{\chi^{++}} - M_{\chi^0}}{2} \times \text{Boost} = \frac{7.00}{2} \times \frac{E_{\chi^{++}}}{M_{\chi^{++}}} \approx 15.8 \text{ GeV}. \quad (75)$$

A measurement of E_ℓ^{max} to $\delta E_\ell^{\text{max}} = 0.1 \text{ GeV}$ translates to $\delta r_{21}/r_{21} = 2\delta E_\ell^{\text{max}}/(M_{\chi^{++}} - M_{\chi^0}) = 2(0.1)/7.00 \approx 0.29\%$, below the stated theoretical uncertainty $\pm 0.3\%$.

C. FCC-hh: $pp \rightarrow \chi^0 \chi^0 + j$

The partonic cross-section for $q\bar{q} \rightarrow \chi^0 \chi^0$ via s -channel Z' at $\hat{s}$:

$$\hat{\sigma}(q\bar{q} \rightarrow \chi^0 \chi^0) = \frac{g_f^2 g_{\text{dark}}^2}{12\pi \hat{s}} \left(1 + \frac{2M_{\chi^0}^2}{\hat{s}} \right) \sqrt{1 - \frac{4M_{\chi^0}^2}{\hat{s}}}. \quad (76)$$

Convoluting with CT18 PDFs at $\mu_F = 2M_{\chi^0}$ and applying the NLO K -factor $K \approx 1.3$ [31]:

$$\sigma(pp \rightarrow \chi^0 \chi^0) = K \cdot \sum_q \int_0^1 dx_1 \int_0^1 dx_2 f_q(x_1, \mu_F) f_{\bar{q}}(x_2, \mu_F) \hat{\sigma}(\hat{s} = x_1 x_2 s) \Theta(\hat{s} - 4M_{\chi^0}^2) \approx 0.8 \text{ fb}. \quad (77)$$

At 30 ab^{-1} : $N_{\text{sig}} \approx 0.8 \times 3 \times 10^7 \approx 2.4 \times 10^4$, giving $\text{SNR} \approx 12\sigma$ over the $Z(\rightarrow \nu\nu) + j$ background in the monojet channel.

TABLE V. Complete summary of numerical results derived in this Supplemental Material. All entries reproducible via <https://github.com/ahmed19999520-alt/dark-octet-dm>. Uncertainties on mass ratios propagate from $\pm 5\%$ on symmetry-breaking coefficients; uncertainties on σ_T/m include $\pm 20\%$ from Sommerfeld numerics; GW SNR uncertainty is $\pm 10\%$ from phase-transition parameters.

Quantity	Symbol	Value	Reference
<i>Mass spectrum</i>			
χ^0 (DM) mass	M_{χ^0}	165.67 GeV	Eq. (29)
Mass ratio r_{21}	$M(\chi^{++})/M(\chi^0)$	1.042 ± 0.003	Eq. (35)
Mass ratio r_{31}	$M(\Sigma_d^0)/M(\chi^0)$	0.809 ± 0.004	Eq. (36)
<i>Relic abundance</i>			
Freeze-out x_f	M_0/T_f	24.6 ± 0.3	Table III
Effective d.o.f.	$g_{\text{eff}}(x_f)$	10.84	Eq. (48)
Relic abundance	$\Omega_{\text{DM}} h^2$	0.120 ± 0.003	Eq. (50)
<i>Self-interactions</i>			
σ_T/m at $v = 10$ km/s	—	$0.52 \pm 0.09 \text{ cm}^2 \text{ g}^{-1}$	Eq. (60)
σ_T/m at $v = 1000$ km/s	—	$0.0042 \pm 0.0009 \text{ cm}^2 \text{ g}^{-1}$	Table IV
Sommerfeld factor (Coulomb)	S^{Coul}	3770 (at 10 km/s)	Eq. (58)
Sommerfeld factor (Yukawa)	S^{Yuk}	1964 (at 10 km/s)	Eq. (59)
Yukawa suppression	$S^{\text{Yuk}}/S^{\text{Coul}}$	0.521	Eq. (59)
<i>Direct detection</i>			
σ_{SI} (kinetic mixing)	$\sigma_{\text{SI}}^{\text{KM}}$	$3.1 \times 10^{-48} \text{ cm}^2$	Main text Eq. (9)
LZ 2024 limit at 165 GeV	—	$5 \times 10^{-47} \text{ cm}^2$	[25]
DARWIN projected sensitivity	—	$3 \times 10^{-49} \text{ cm}^2$	[26]
<i>Gravitational waves</i>			
Peak frequency	f_*	$1.65 \times 10^{-3} \text{ Hz}$	Eq. (67)
Peak amplitude	$\Omega_{\text{GW}}^{\text{peak}}$	$\sim 10^{-8}$	Eq. (63)
LISA SNR (4 yr)	SNR	8.5 ± 0.9	Eq. (69)
<i>Collider</i>			
$\sigma(e^+e^- \rightarrow \chi^+\chi^-)$ at 365 GeV	—	$1.1 \times 10^{-3} \text{ fb}$	Eq. (72)
Events at FCC-ee (1.5 ab^{-1})	N_{sig}	≈ 495	Eq. (73)
$S/\sqrt{B}$ at FCC-ee	—	$\approx 5.3\sigma$	Sec. VII
$\sigma(pp \rightarrow \chi^0\chi^0)$ at 100	—	0.8 fb	Eq. (77)
<i>Topological quantities</i>			
Instanton action	S_{inst}	3948	Eq. (16)
Stability amplitude	$e^{-S_{\text{inst}}}$	$\approx 10^{-1714}$	Eq. (16)
BPS mass bound	$M_{G_M}^{\text{BPS}}$	$1.73 \times 10^{15} \text{ GeV}$	Eq. (22)
Core radius	r_{core}	0.206 GeV^{-1}	Table I

VIII. SUMMARY OF ALL NUMERICAL RESULTS

IX. COMPUTATIONAL REPRODUCIBILITY

All calculations presented in this Supplemental Material are implemented in the public repository

<https://github.com/ahmed19999520-alt/dark-octet-dm>

with the following module-to-calculation correspondence:

The code environment is reproducible via Docker:

```
docker compose -f docker/docker-compose.yml up
# Open http://localhost:8888
```

The test suite (847 unit tests) verifies every numerical claim in this document:

```
pytest tests/ -v --cov=src/dark_octet
```

TABLE VI. Repository module correspondence.

Module	Content
<code>src/dark.octet/homotopy/</code>	Sections I–II
<code>src/dark.octet/spectrum/</code>	Section III
<code>src/dark.octet/boltzmann/</code>	Section IV
<code>src/dark.octet/sommerfeld/</code>	Section V
<code>src/dark.octet/signatures/</code>	Sections VI–VII
<code>notebooks/AppendixA.Homotopy.ipynb</code>	Section I
<code>notebooks/AppendixB.Boltzmann.ipynb</code>	Section IV
<code>notebooks/AppendixC.Sommerfeld.ipynb</code>	Section V

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